Mathematical overview: who is left standing

The game and its assumptions. People $1, \ldots, n$ stand in a circle (counters on spots 1 to n). Starting at 1, the people still standing take turns saying a repeating chant: "stand, sit" (every second person sits) or "stand, stand, sit" (every third). Whoever gets "sit" sits down and is skipped from then on; empty spots are skipped too. The chant carries on past 1 without starting over. Write $J(n)$ for the last one standing when every second person sits and $J_3(n)$ when every third sits. This is the Josephus problem; everything below is proved for all n.

Step rule, for either chant. For $n \geq 2$, adding one person moves the answer up by the length of the chant, wrapping round: $J(n) = J(n-1) + 2$ and $J_3(n) = J_3(n-1) + 3$, minus n whenever the sum is more than n. *Proof.* The first person to sit is the one who gets the last word of the chant: 2, or 3 (person 1 in a circle of 2 with the three-word chant). The next standing person gets the first word, so the rest of the game is exactly a circle of $n-1$ whose "person 1" is old person 3 (or 4), counting round. Every number in that smaller circle is 2 (or 3) less than in the big one, so translating its answer back gives the rule.

Every second person: if $n = 2^m + L$ with $0 \leq L < 2^m$, then $J(n) = 2L + 1$. *Proof.* (i) In a circle of 2^m where the first person to count stays, that person is last: a lap round an even circle seats every second person and ends on a "sit", so the next lap again starts with the same person staying, in a circle half the size. (ii) In a circle of $2^m + L$, the first L people to sit are $2, 4, \ldots, 2L$, all in the first lap since $2L < n$. Then 2^m people remain and the next word, a "stand", goes to $2L + 1$; by (i), $2L + 1$ is last.

Consequences. $J(n) = 1$ exactly when n is a power of two, and $J(n) = n$ exactly when n is one less than a power of two $(1, 3, 7, 15, 31, \ldots)$; between powers of two the answers climb $1, 3, 5, \ldots$ (the step rule: up by 2, back to 1 when $J(n-1) = n - 1$). No even number is ever last, because every even number sits in the first lap, while each odd number $2L + 1$ is last in the circle of $2^m + L$ for every power of two $2^m > L$ (9 is last in 12, 20, 36, $\ldots$). Doubling: $J(2k) = 2J(k) - 1$ and $J(2k + 1) = 2J(k) + 1$, since after the first lap of $2k$ people $1, 3, \ldots, 2k - 1$ remain with 1 next, while in a circle of $2k + 1$ the lap ends with $2k + 1$ standing and 1 sitting, leaving $3, 5, \ldots, 2k + 1$ with 3 next. Binary: n is a 1 followed by the digits of L, and $2L + 1$ is the digits of L followed by a 1, so the answer moves the leading 1 of n to the end ($12 = 1100 \to 1001 = 9$; $8 = 1000 \to 0001 = 1$).

Every third person: a step rule, no simple formula. For $n = 1$ to 12, $J_3(n)$ is 1, 2, 2, 1, 4, 1, 4, 7, 1, 4, 7, 10; $J_3(30) = 29$ and $J_3(41) = 31$. The step rule gives each from the one before; there is no comparably simple formula, and the pages ask only for the step rule. Multiples of 3 are never last, since they all sit in the first lap. Where to stand to be last (K–1 Problem 7, grades 2–3 and 4–5 Problem 9) means finding $J_3(n)$.

Limits, and the one misreading that matters. Counting starts at person 1, who stays; starting elsewhere only renames everyone. A child who restarts the chant at 1 on every lap never seats 1 and finds 1 last every time. Circles of 2, 4, 8 and 16 cannot reveal this, since every lap there really does begin at 1 with "stand"; the circle of 3 is the first that does (2 sits, 3 stands, 1 sits).

What children discover: experiment, conjecture, explanation. Playing circles and filling tables are experiments. "1 is last at 1, 2, 4, 8, 16", "up by 2, back to 1 at a power of two" and "add 3 and wrap" are conjectures read from tables. Explanations within reach: "2 always sits first" and "every even number sits the first time round" (all bands, with counters); the halving laps for 16 and 32 (2–3 and 4–5, on the small ring); "once L people have sat, a power of two are left and the next person stays" (4–5, and some third graders); the renaming from 26 to 52 and 53, and the binary rule (4–5). The step rules' proof is for 4–5 children who ask why.

Band map and prerequisites. K–1 (count and recognize numerals to 16; the adult reads) plays circles of 3 to 10 and 16 on the printed rings, guessing first; finds that 4, 8 and 16 leave 1 and which circles leave 3 and 5; and explains why 2 can never be last (on a return visit, why 3 can never be last with stand, stand, sit). Grades 2–3 (an adult reads the opening rules; problems are one or two sentences; adding 2s and 3s to 100) build the table to 16, predict 20, 25 and 32, explain the even numbers and the halving for 16 and 32, search up to 40, reach 50 and 100, and extend an every-third table to 30. Grades 4–5 (doubling; binary is introduced on the page) explain powers of two and $2L + 1$ for 100 and 1000, use doubling, show that 22 is never last, extend every third to 41 and explain the binary rule.

Preparation, launch and pacing

Status. Prepared for organizer review; unpiloted. Not physically tried: 25 mm (1 inch) counters on the 26 mm printed spots, where neighbouring spots are as little as 2 mm apart (the K–1 rings of 9 and 10 and the 16-spot rings); the two halves of the 32-spot mat laid side by side; how long pairs keep an accurate count; the launch. All timings are estimates, and that K–1 pairs can chant and flip accurately up to the circle of 16 is a prediction; if they cannot, the adult chants while the children point and flip.

Materials and printing. Two-colour counters 25 mm (1 inch) across and no wider, 16 per K–1 pair and 32 per older pair; one colour up means standing, and a counter is flipped when that person sits. Pencils; optionally eleven number cards for the launch. Print US Letter, single-sided, at 100% (Actual size), so the spots stay 26 mm. Nothing is cut.

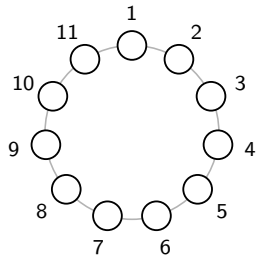
- K–1 table (two pairs): 32 counters; one 8-page packet per child (32 sheets). Every page except page 5 is a working board: counters go straight onto the printed rings, and page 4's 16-spot ring is full size.
- Grades 2–3 table (two pairs): 64 counters; pages 1–4 per child, and pages 5–7 (the 16-spot mat and the two halves of the 32-spot mat) once per pair (22 sheets).
- Grades 4–5 table (a pair and a referee): 32 counters; pages 1–5 per child, and pages 6–8 once (18 sheets).

Lay the 32-spot halves side by side, spot 1 at the top left and inner edges touching, with a strip of tape on the back; the joined mat is about 43 by 28 cm. Beyond printing, preparation is about 15 minutes: count counters into five bags (two of 16, three of 32) and tape three 32-spot mats. In all: 128 counters and 72 sheets. Plain 25 mm counters with a sticker on one side also work (add about 20 minutes).

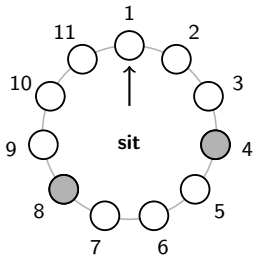
Staffing. One adult at each table, as usual. In each pair one child touches each standing counter and says its word, and the partner flips on "sit"; they swap every circle. The 4–5 third child referees, watching the finger and the wrap past 1, and rotates in. The adult's main job is the wrap: watch each pair's first circle that passes 1. Adults read problems aloud and may keep a pooled record; children make the guesses, predictions and explanations. If the adult is doing the counting, move to a smaller circle.

How the K–1 table runs. A pair plays each circle once, on one child's page (take turns whose), one ring at a time: a counter on every spot of that ring, play, and move the counters to the next ring after drawing the ring round the last number. Pages 1, 3 and 6 have more spots than the 16 counters, so never fill a whole page. Both children draw the ring on their own page. In Problem 4 children point to the rings; the adult writes each circle's size (its biggest number) in the boxes, unless a child wants to, and scribes spoken reasons for Problems 5 and 8. That is routine recording; the children still make every guess and choice.

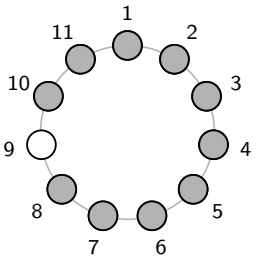
Launch, about 5 minutes after running around. All eleven children stand in one circle and an adult numbers them 1 to 11 clockwise. Use a chant that no problem uses: walking round and touching each standing child, the adult says "stand, stand, stand, sit". On "sit" that child sits on the floor; seated children are skipped. Pause after 9, 10 and 11 have each had "stand" and ask who gets the next word. The point to make: the chant does not start over at 1, so person 1 gets "sit". Play on until one child is standing. Any circle of 7, 11 or 15 works (one less than a multiple of 4), because person 1 then gets "sit" right at the wrap; add adults to reach one, and a child can be the chanter. (The last one standing is 2 in a circle of 7 and 13 in a circle of 15.) If a picture helps, draw this on the whiteboard:



11 standing; count from 1



4 and 8 sat; after 9, 10, 11, 1 gets "sit"



everyone else sat; 9 is last

The sitting order is 4, 8, 1, 6, 11, 7, 3, 2, 5, 10. The launch avoids every-second-person circles because each circle up to 16 is a table entry on the older pages.

At the tables: the same start on counters. Hold up a counter: on the mats a counter is a person, one side up for standing, flipped when that person sits. Children handle the counters for a minute. Then each adult replays the start of the launch on the table's own materials, touching each standing counter and saying "stand, stand, stand, sit" while a child flips (about 30 seconds):

- Grades 2–3 and 4–5: put 11 counters on spots 1–11 of the 16-spot mat (2–3 page 5, 4–5 page 6). 4 sits, 8 sits, then 9, 10 and 11 stand, the chant skips the empty spots 12–16, and 1 sits. Stop and reset.
- K–1: on the ring of 7 (page 2), 4 sits, then 5, 6 and 7 stand, and 1 sits. Stop and reset. Use this ring rather than page 4's, because the K–1 rule is a counter on every spot.

Then give the table's own chant, "stand, sit", and show where a result goes: a ring round the last one's number (K–1) or an entry in the Problem 1 table (2–3 and 4–5).

A first visit. About 35–40 minutes at the tables, then a few minutes of sharing.

- K–1: Problems 1–4, with Problem 5 as the closing conversation (about 30 minutes, playing each circle once: 59 flips, 15 of them in the circle of 16). Problems 6–8 (stand, stand, sit) are a return visit, or for a pair that finishes.
- Grades 2–3: Problems 1 and 2 take about 15–20 minutes. Problem 1 is shared between the pairs (one plays 1 to 11, the other 12 to 16: 55 and 65 flips; results pooled in one table on a sheet or whiteboard, and each pair replays two of the other's circles); in Problem 2 each pair plays one or two of the 32-spot circles. Problems 3 and 4 fill the rest of the hour. Return: Problems 5–6, then 7–9. A pair that stalls on 5 or 6 can move to 7, which is concrete again.
- Grades 4–5: Problems 1 and 2, with roles rotating each circle (two play, one referees), take about 15–20 minutes: 194 flips for the one playing pair. Then Problems 3 and 4, and 5–6 if the table is quick. Return: 5–6, then 7–9. Problem 10 is the deep end and needs binary; save it for a visit when the table is fresh.
- Sharing: each table reports one finding, such as which circles left 1, who is last among 100, or why 22 can never be last.

Hints only when useful. Counting slips are the main risk, and one wrong entry hides a pattern. When an entry breaks a pattern, ask the pair to replay that circle aloud with the partner watching, rather than correcting the entry. If a child restarts the chant at 1, ask what word the person before 1 got and what word comes next. If someone gives a word to a sitting counter or an empty spot, ask whether a sitting person gets a turn.

Grades K–1: solutions and hints

Plain-words key for the adult. Stand, sit: circles of 3, 4, 5, 6 leave 3, 1, 3, 5; circles of 7, 8, 9, 10 leave 7, 1, 3, 5; the circle of 16 leaves 1. Stand, stand, sit: circles of 3 to 8 leave 2, 1, 4, 1, 4, 7, and the circle of 9 leaves 1. Use the key to know when to ask for a replay, not to give answers.

Problem 1, page 1: circles of 3 to 6

3, 1, 3, 5. Circle of 3: 1 stands, 2 sits, 3 stands, then 1 sits, so 3 is last. Circle of 6: 2, 4, 6 sit; then 1 stands, 3 sits, 5 stands, 1 sits, and 5 is last. A pair that finds 1 last in the circle of 3 has started over at 1. Hint: "What word did 3 get? What word comes after stand?"

Problem 2, pages 2–3: guess, then play 7 to 10

7, 1, 3, 5. Guesses are spoken; any guess is fine, but ask what it is based on. Children may notice that the last one is never 2, or that the circle of 8 went like the circle of 4. Hint: "Which circle that you already played is most like this one?"

Problem 3, page 4: stars, and the circle of 16

Stars on the circles of 4 (page 1) and 8 (page 2), and on the circle of 16 once it is played: it leaves 1. A child who says "4, 8, then 16" or "it keeps doubling" has a real conjecture, and children watching the counters may see why: after the first time round, half are sitting and 1 starts again. Hint: "Look at the starred circles. What

comes after 4 and 8?" Extension, spoken: which bigger circle would leave 1? (32.)

Problem 4, page 5: which circles left 3, which left 5

Number 3: circles of 3, 5 and 9. Number 5: circles of 6 and 10. These lists are complete for the circles on the pages; children find them on their rings. Hint: "Where did you draw a ring round a 3?" Extension, spoken: "Which circle comes just after a star?" Number 3 comes round again just after each star (5 after 4, 9 after 8), and 5 one circle later; bigger circles 17 and 33 also leave 3, and 18 and 34 leave 5.

Problem 5, page 5: can 2 be last?

No. Number 2 always gets the first "sit", in every circle that has a 2. "It never happened" is evidence, not yet a reason; the follow-up "Could it happen in a circle we haven't played?" invites the reason. Hint: "In your circles, when did 2 sit down?" Extension: can 4 be last? (No: 4 gets the second "sit"; every even number sits the first time round.)

Problem 6, pages 6–7: stand, stand, sit, circles of 3 to 8

2, 1, 4, 1, 4, 7. Circle of 3: 3 sits; then 1 and 2 stand and 1 sits, so 2 is last. Circle of 4: 3 sits; 4 and 1 stand, 2 sits; 4 and 1 stand, 4 sits; 1 is last. The three-word chant across the wrap is harder than before, so the finger-and-flip roles matter more. Hint: "Where is your finger now, and who gets the next word?"

Problem 7, page 8: where to stand in the circle of 9

At 1. The sitting order is 3, 6, 9, 4, 8, 5, 2, 7. Choosing where to stand is the same as finding who is last; children may guess, then play to find out. Hint: "How could you find out without guessing?"

Problem 8, page 8: can 3 be last with stand, stand, sit?

No. Number 3 always gets the first "sit". Hint: "In your stand, stand, sit circles, when did 3 sit down?" Extension: can 6 be last? (No: 6 gets the second "sit".)

Grades 2–3: solutions and hints

Problem 1, page 1: circles of 1 to 16

1, 1, 3, 1, 3, 5, 7, 1, 3, 5, 7, 9, 11, 13, 15, 1. The circle of 1 has nobody to sit. The pattern: a 1 at 1, 2, 4, 8 and 16, with the odd numbers climbing by 2 in between. Hint: "Where are the 1s? What happens just after each one?"

Problem 2, page 1: predict 20, 25, 32

9, 19, 1. Continuing the climb after 16: 17 leaves 3, . . . , 20 leaves 9, 25 leaves 19, 31 leaves 31, and 32 leaves 1. Predictions go in the table before playing. Each check takes 19 to 31 sits on the 32-spot mat; if a play disagrees with a prediction, replay before deciding which is wrong. Hint: "What did your table do after 8? What might it do after 16?"

Problem 3, page 1: numbers that can never be last

Exactly the even numbers. The first time round everyone is standing, so the words go stand, sit, stand, sit in order, and every even number sits before the chant reaches 1 again. Every odd number does win somewhere: each climb after a power of two runs through all the odd numbers below the next power of two (17 is last in the circle of 24). For a child who answers from the table alone: "Could an even number win in a circle bigger than 16?"

Problem 4, page 2: why 16 and 32 leave 1

In the circle of 16, the first lap seats 2, 4, . . . , 16. Number 16 got the lap's last word, a "sit", so 1 gets "stand" again: it is now a circle of 8 starting at 1. The next laps seat 3, 7, 11, 15, then 5, 13, then 9. Each lap halves the circle and ends on a "sit", so 1 is never told to sit. After the first lap of 32, the 16 odd numbers remain with 1 next, which is the circle of 16 again. On the small ring children can cross out one lap at a time. Hints: "After the first time round, how many are standing, and who says the next word?" "Have you played a circle like that?"

Problem 5, page 2: number 9, and the same number

Number 9: circles of 12, 20 and 36. Same number: 1, 3, 7, 15, 31 (accept with or without 1). The lists are complete because 9 appears once in each climb that reaches it, four circles after the power of two 8, 16 or 32; the climb after 4 stops at 7, and the one after 64 starts beyond 40. The last one equals the circle size at the top of each climb, just before a power of two. Hint: "Where does 9 come in each climb?" Discretionary follow-up: "How do you know there are no more up to 40?"

Problem 6, page 3: circles of 50 and 100

37 and 73. The nearest power of two below 50 is 32, which leaves 1; each extra person adds 2, and 50 is 18 more, so $1 + 2 \times 18 = 37$. Another reason: once 2, 4, ..., 36 have sat (18 people), 32 remain and 37 is next, a fresh circle of 32 starting at 37. For 100: $100 = 64 + 36$, so $1 + 72 = 73$. Hints: "Which circle near 50 are you sure about?" "What happens to the answer when one more person joins?" Why the climb goes up by 2 is the step rule in the overview.

Problem 7, page 3: every third, circles of 1 to 12

1, 2, 2, 1, 4, 1, 4, 7, 1, 4, 7, 10. In the circle of 2 the words go 1 stand, 2 stand, 1 sit, so 2 is last. Hint for small circles: "What if you keep going round, even with only two people?"

Problem 8, page 4: predict 13 to 20, check 14 and 20

13, 2, 5, 8, 11, 14, 17, 20; the checks give 2 and 20 (20 needs the 32-spot mat). The pattern: add 3 to the previous answer, and if that is more than the circle size, take the circle size away ($10 + 3 = 13$; $13 + 3 = 16$ and $16 - 14 = 2$). The 1–12 table drops to 1 at 4, 6 and 9, so a child may predict "back to 1"; the circle of 14 lands on 2, which is why it is a check. Hint: "How much does it go up each time? What happens when it would go past the circle size?"

Problem 9, page 4: where to stand among 30

At 29. Continuing from 20, circles of 21 to 30 leave 2, 5, 8, 11, 14, 17, 20, 23, 26, 29. Playing 30 on the 32-spot mat is a long check, not the main route. Hint: "Can your table keep going past 20?" Extension: Grades 4–5 Problem 9 (41 people; stand at 31).

Grades 4–5: solutions and hints

Problems 1 and 2, page 1

As Grades 2–3 Problems 1 and 2: the table 1, 1, 3, 1, 3, 5, 7, 1, 3, 5, 7, 9, 11, 13, 15, 1, and 20, 25, 32 leave 9, 19, 1.

Problem 3, page 1: every circle up to 100 that leaves 1

1, 2, 4, 8, 16, 32, 64 (accept without 1). The first lap explains both directions. In an even circle the first lap seats the even numbers and 1 starts again in a circle half the size; in an odd circle bigger than 1, the last person of the first lap stands and then 1 sits. So 1 is last exactly when halving never meets an odd number above 1, which means a power of two. The small ring is for crossing out the laps of 16. Hint: "After the first lap, how many are left, and who goes next?" Discretionary follow-up: "How do you know no other circle up to 100 leaves 1?"

Problem 4, page 2: circles of 100 and 1000

73 and 977. $100 = 64 + 36$: once 36 people have sat (2, 4, ..., 72), 64 remain and 73 is next, so by Problem 3's reasoning 73 is last. $1000 = 512 + 488$: after 2, 4, ..., 976 sit, 977 starts a circle of 512. Hint: "Which power of two is just below 100? How many must sit to get down to it, and who is next?"

Problem 5, page 2: from 26 to 52 and 53

41 and 43. In the circle of 52, after the first lap $1, 3, \ldots, 51$ remain with 1 next. That is the circle of 26 with person j renamed $2j - 1$, so the 21st is $2 \times 21 - 1 = 41$. In the circle of 53, the first lap seats 2 to 52, then 53 stands and 1 sits; $3, 5, \ldots, 53$ remain with 3 next, person j renamed $2j + 1$, so the answer is $2 \times 21 + 1 = 43$.

Hint: "After the first lap of 52, who is still standing, and who goes next? What circle does that look like?" Extension: the general doubling rule, and the chain 25, 50, 100 (19, 37, 73) as a check on Problem 4.

Problem 6, page 3: number 23, number 22

Number 23: circles of 27, 43 and 75. Number 22: none, because every even number sits in the first lap. The 23 list is complete because $23 = 2 \times 11 + 1$: the circle must be a power of two plus 11, with the power bigger than 11, so $16 + 11$, $32 + 11$ and $64 + 11$ ($8 + 11$ fails, since the climb after 8 stops at 15, and $128 + 11$ is over 100). In children's terms, 23 comes 11 circles after a power of two, if that climb is long enough. Hints: "How many circles after 16 does the climb reach 23?" For 22: "When does 22 sit down?"

Problem 7, page 3: every third, circles of 1 to 12

As Grades 2–3 Problem 7: 1, 2, 2, 1, 4, 1, 4, 7, 1, 4, 7, 10.

Problem 8, page 4: predict 13 to 24, check 14 and 21

13, 2, 5, 8, 11, 14, 17, 20, 2, 5, 8, 11; both checks give 2 (21 needs the 32-spot mat). Both land just past a wrap, so they test "add 3, and take away the circle size" against "drop to 1". For a child who asks why the step works: once 3 sits, 4 starts a circle of $n - 1$ (the proof is the step rule's, in the overview). Hint: "How does each answer compare with the one before?"

Problem 9, page 4: where to stand among 41

At 31. Continuing the step rule from 24: circles of 25 to 30 leave 14, 17, 20, 23, 26, 29; the circle of 31 leaves 1; then 4, 7, 10, ..., and the circle of 41 leaves 31. Number 16 is the last to sit, so 16 and 31 are the last two standing. Hint: "Do you need to play 41, or can the table take you there?"

Problem 10, page 5: the binary rule

Move the leading 1 to the end, dropping any zeros that end up in front: $13 = 1101 \to 1011 = 11$, $25 = 11001 \to 10011 = 19$, $16 = 10000 \to 00001 = 1$. Why: the leading 1 stands for the largest power of two 2^m in n, and the remaining digits are L. Moving the 1 to the end shifts L's digits up one place, which doubles it, and adds 1, giving $2L + 1$, the rule behind Problem 4's answers. The printed example rules out "reverse the digits" ($1100 \to 0011 = 3$). "Swap the second and last digits" fits 12 but fails at once on 8 and 16. Hint: "Try your rule on 8 and on 16. What should it give?" If binary is new: "Which of 32, 16, 8, 4, 2, 1 add up to 25?"

Sources, evidence and what to record

Ronald L. Graham, Donald E. Knuth and Oren Patashnik, *Concrete Mathematics*, 2nd ed. (Addison-Wesley, 1994), §1.3, "The Josephus Problem", which the outline cites for the recurrences and the binary rotation. Not consulted for this guide (no copy is in the project's references); page numbers unchecked. The proofs above are written independently.

Year 1, Handouts 6.7 and 7.8 (`lowell-math-circle-year-1`): consulted. They ask how to order a deck so that a deal gives $1, 2, \ldots, n$, but do not state the deal. If it moves every other card to the bottom, it is this elimination run backward; confirm with the organizer before calling this week a return.

Evidence: `check_guide.py` (in this guide's source folder) simulates the game literally, confirms every statement in the overview for every second person up to $n = 2048$ and every third up to 500, and recomputes every answer, list and count printed here, including the launch and its replay on counters, the flip counts behind the timings, and the spot spacing. These are digital checks; nothing here has been rehearsed with counters or children.

Library connections, from the outline: Week 4 hops a fixed number of dots round a ring, Week 89 grows patterns layer by layer, and Weeks 8 and 30 use binary and ternary. New here: an elimination process on a circle with a doubling recurrence and a binary rule.

Record after a session: preparation time; whether the counters fit the spots and the mat halves stayed together; whether pairs kept the chant going past 1 without reminders; which entries were wrong and how they were caught; which problems each table reached; and what children wanted to continue.